

BTECCE23501:: Operating Systems

Course Type	PCC	Semester	5
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	--Hr./Week	Practical/Studio	-	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course helps the student to identify the role of operating systems and explain the different structures of operating systems. It highlights the OS support for processes and threads, depicts scheduling algorithms and deadlock mechanisms. It also identifies virtual memory requirement of an operating system and illustrates the importance of I/O and file systems.

Course Outcomes

CO No.	Statement
1	Examine the functions of a contemporary Operating system with respect to convenience, efficiency and the ability to evolve.
2	Demonstrate knowledge in applying system software and tools available in modern operating system, such as threads, system calls, semaphores for software development.
3	Apply various CPU scheduling algorithms and identify deadlock mechanisms to construct solutions to real world problems.
4	Understand the organization of memory and memory management hardware.
5	Analyze I/O and file management techniques for better utilization of secondary memory.

Mapping of COs to POs and PSOs

CO No	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1	2																UN
CO2		2															AN
CO3						3											AP
CO4							3										AN
CO5															3		AP

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction	Hours 9	CO	BTL
	Introduction to OS: What is OS, Interaction of OS and hardware, Goals of OS, Basic functions of OS, OS Services, System Calls. Types of OS: Batch, Multiprogramming, Time sharing, Parallel, Distributed & Real-time OS. Real Time OS: Concepts, Design Issues, Architecture, Real Time Kernel Embedded Systems: Definition, Classification, Characteristics, Components: Microcontrollers, sensors, actuators, communication modules, and memory. Embedded System Design cycle. Virtualization technologies: Hypervisor Containerization platforms: Docker, Kubernetes Shell: Linux commands and shells, shell programming, AWK programming. Introduction to Mobile OS: Architecture & Overview of Android OS.		1	UN
Unit No 2	Process Management	Hours 9	2	AN
	Processes: Process Concept, Process State Model, Process Description Threads: Multithreading models, Symmetric Multiprocessing, User level and kernel level threads, Microkernels. Concurrency: Issues with concurrency, Principles of Concurrency Mutual Exclusion: H/W approaches, S/W approach, OS/Programming Language support: Semaphores, Mutex, Monitors. Classical Problems of Synchronization: Readers-Writers problem, Producer Consumer problem, Dining Philosopher problem.			
Unit No 3	CPU Scheduling and Deadlock	Hours 9	3	AP
	Uniprocessor Scheduling: Types of Scheduling - Preemptive, Non-preemptive, Long-term, Medium-term, Short-term, Scheduling Criteria. Scheduling Algorithms: FCFS, SJF, RR, Virtual Round Robin, Priority Real time Scheduling Algorithms: Earliest Deadline First, Rate Monotonic Scheduling, Least Laxity First Deadlock: Principles of deadlock, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Deadlock Recovery. Case studies: Scheduling in LINUX and Windows Case studies: Deadlock management in LINUX and Windows			
Unit No 4	Memory Management	Hours 9	4	AN
	Memory Management concepts: Memory Management requirements Memory Partitioning: Fixed, Dynamic Partitioning, Buddy Systems. Placement Strategies: First Fit, Best Fit, and Worst Fit. Fragmentation, Swapping, Thrashing. Virtual Memory: Concepts, Address translation, VM with Paging, Page Table Structure, Translation Look-aside Buffer, VM with Segmentation, Working Set Page Replacement Policies: FIFO, LRU, Optimal.			

Case studies: Memory management implementation in LINUX and Windows				
Unit No 5	I/O & Device Management	Hours 9	5	AP
I/O management: I/O Devices - Types, Characteristics of Serial and Parallel devices, OS design issues for I/O management, I/O Buffering. Disk Scheduling: FCFS, SCAN, C-SCAN, SSTF. File Management: Concepts, File Organization, File Directories, File Sharing. Record Blocking, Secondary Storage Management, Free Space management, Security. File System: Structure, Implementation, Memory mapped files, Study of LINUX File system				

Textbooks

1	<i>“Operating Systems”, Stallings William, Pearson Education, ISBN: 0-13-031999-6, 5th Edition.</i>
2	<i>“Operating System Concepts”, Silberschatz A., Galvin P., Gagne G., John Wiley and Sons, ISBN: 9971-51-388-9, 6th Edition.</i>

Reference Books / Journal Articles / Weblink

1	<i>“Modern Operating Systems”, Tanenbaum Andrew S., Prentice Hall India, ISBN: 81-7808-447-3, 2nd Edition.</i>
2	<i>“Unix Concepts and Applications”, Das Sumitabha, Tata McGraw Hill, ISBN: 0-07-053475-6, 3rd Edition.</i>
3	<i>“Unix Shell Programming”, Yashavant Kanetkar, BPB Publications, ISBN: 81-7029-753-2, 1st Edition.</i>
4	<i>"Embedded systems: Architecture, programming and design", Rajkamal, McGraw-Hill, ISBN: 978-0070588440, 2003.</i>
5	<i>“Real-time systems”, Jane W. S. Liu, Prentice Hall, 2000.</i>